

```

1  #include <stdio.h>
2  #include <string.h>
3  #include <ctype.h>
4
5  #define MAX_KEYWORDS 32
6  #define MAX_WORD_LEN 20
7
8  struct keyword {
9      char word[MAX_WORD_LEN];
10     int count;
11 };
12
13 int isKeyword(char word[]) {
14     char *keywords[MAX_KEYWORDS] = {
15         "auto", "break", "case", "char", "const", "continue", "default",
16         "do", "double", "else", "enum", "extern", "float", "for", "goto",
17         "if", "int", "long", "register", "return", "short", "signed",
18         "sizeof", "static", "struct", "switch", "typedef", "union",
19         "unsigned", "void", "volatile", "while"
20     };
21
22     for (int i = 0; i < MAX_KEYWORDS; i++) {
23         if (strcmp(word, keywords[i]) == 0)
24             return 1;
25     }
26     return 0;
27 }
28
29 int findKeyword(struct keyword k[], int count, char word[]) {
30     for (int i = 0; i < count; i++) {
31         if (strcmp(k[i].word, word) == 0)
32             return i;
33     }
34     return -1;
35 }
36
37 int main() {
38     FILE *fp;
39     char ch, word[MAX_WORD_LEN];
40     struct keyword found[MAX_KEYWORDS];
41     int i = 0, kcount = 0;
42
43     fp = fopen("input.c", "r");
44     if (fp == NULL) {
45         printf("Error: File not found!\n");
46         return 0;
47     }
48
49     while ((ch = fgetc(fp)) != EOF) {
50         if (isalpha(ch)) {
51             i = 0;
52             while (isalpha(ch)) {
53                 word[i++] = ch;
54                 ch = fgetc(fp);
55             }
56             word[i] = '\0';
57
58             if (isKeyword(word)) {
59                 int index = findKeyword(found, kcount, word);
60                 if (index == -1 && kcount < MAX_KEYWORDS) {
61                     strcpy(found[kcount].word, word);
62                     found[kcount].count = 1;
63                     kcount++;
64                 } else if (index != -1) {
65                     found[index].count++;
66                 }

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67     }
68 }
69 }
70
71 fclose(fp);
72
73 printf("\nKeyword\t\tCount\n");
74 printf("-----\n");
75 for (i = 0; i < kcount; i++) {
76     printf("%-10s\t%d\n", found[i].word, found[i].count);
77 }
78
79 return 0;
80 }
```